Evaluating Cryptographic API Misuse Detectors for Go

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A Cryptographic API Misuse

Traefik, CVE-2025-66491. ingress-nginx provider:

```
InsecureSkipVerify: strings.ToLower(
    ptr.Deref(cfg.ProxySSLVerify, "off")) ==
        "on",
```

- The user sets `proxy-ssl-verify: "on"`, expecting TLS verification to be **enabled**.
- Expression evaluates to `true`, `InsecureSkipVerify` becomes `true`, inverting security assumptions.
- TLS encryption remains, but backend authentication is disabled: Traefik may trust a spoofed backend.

Prior Work

- **Java.** Comparative studies of crypto misuse detectors[1] show substantial disagreement between tools and frequent developer rejection of alerts.
- **Python.** Evaluations in Python[2] suggest that cryptographic misuse patterns differ across ecosystems, so conclusions from Java do not transfer directly.
- **Go.** Go-specific detectors have been proposed: *CryptoGo*[3] introduced the first approach, and GOPHER [4] extended it. However, they were not evaluated against industry tools.

Gap: no comparative evaluation yet exists for Go.

[1] Zhang et al. 2023, "Automatic Detection ..."; Chen et al. 2024, "Towards Precise Reporting ..."; Ami et al. 2022, "Why Crypto-detectors Fail."
[2] Wickert et al. 2021, "Python Crypto Misuses in the Wild"; Frantz et al. 2024, "Methods and Benchmark ..."; Guo et al. 2024, "CryptoPyt."
[3] Li et al. 2022, "CryptoGo." [4] Zhang et al. 2024, "Gopher."

Problem

Cryptographic API misuse remains a practical security risk in Go systems.

Although several Go detectors already exist (Gopher, CodeQL, Gosec, Snyk Code), their coverage, overlap, and blind spots are still not systematically understood.

Research Questions

- **RQ1.** What is the design space of state-of-the-art tools for detecting cryptographic API misuses in Go?
  - We survey the tools and consolidate a taxonomy of 14 Go crypto misuse classes.
- **RQ2.** To what extent are crypto misuse detection tools for Go consistent with each other?
  - We run all four tools on 328 security-relevant Go projects and compare their findings.

Sourced Tools

CodeQL

GitHub • Open source

Datalog queries over a fact database, with data-flow reasoning.

Gosec

Community • Open source

SSA for some rules, AST pattern matching for others.

Gopher

Academic • Binary-only

gopher

SSA-based taint tracking, with dynamically updatable rules.

Snyk Code

Commercial • Proprietary

snyk

Static analysis product; implementation details undisclosed.

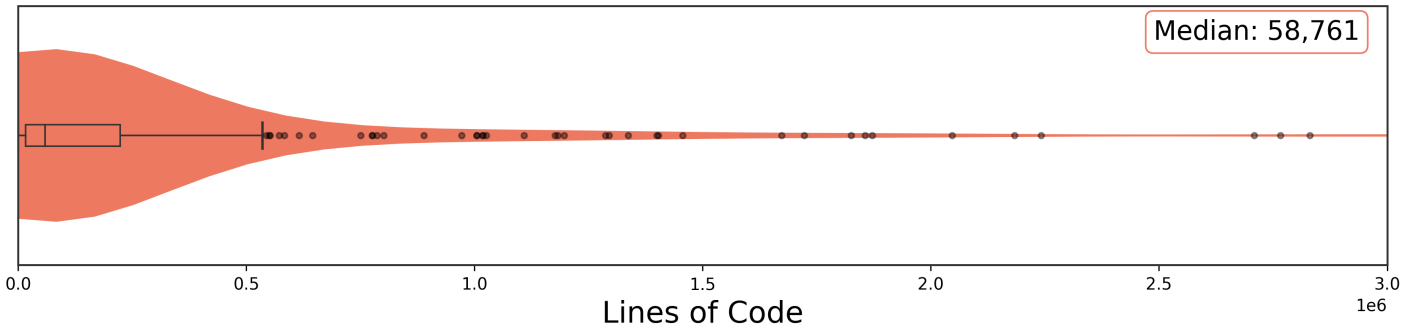
RQ1: Taxonomy and Tool Coverage

Category	#	Rule	CodeQL	Gopher	Gosec	Snyk
Primitives	01	Insecure algorithms		✓	✓	✓
	02	Insecure PRNG	✓	✓	✓	✓
	03	Deprecated Go function		✓		
Key mgmt	04	Constant/predictable key		✓		
	05	Short key length	✓	✓	✓	✓
	06	Static/predictable IV		✓	✓	
PBKDF	07	Short salt length		✓		
	08	Predictable salt		✓		
	09	Low hash iterations		✓		
Transport	10	HTTP protocol		✓		
	11	TLS/SSL issues	✓	✓	✓	✓
SSH	12	Insecure SSH suite		✓		
	13	No host key validation	✓	✓	✓	
Token	14	No JWT verification	✓			

- **Coverage:** Gopher (13) > Gosec (6) > CodeQL (5) > Snyk Code (4)
- **All tools:** 3 of 14 rules (#02, #05, #11) are covered by all

RQ2: Dataset

- **328 open-source Go repositories**, filtered from a curated set of 920 popular projects.
- Kept only actively maintained projects (commit within the past year, stable release) where cryptography is core.
- Size is right-skewed: median 58,761 LOC.



RQ2: Tool Execution and Coverage

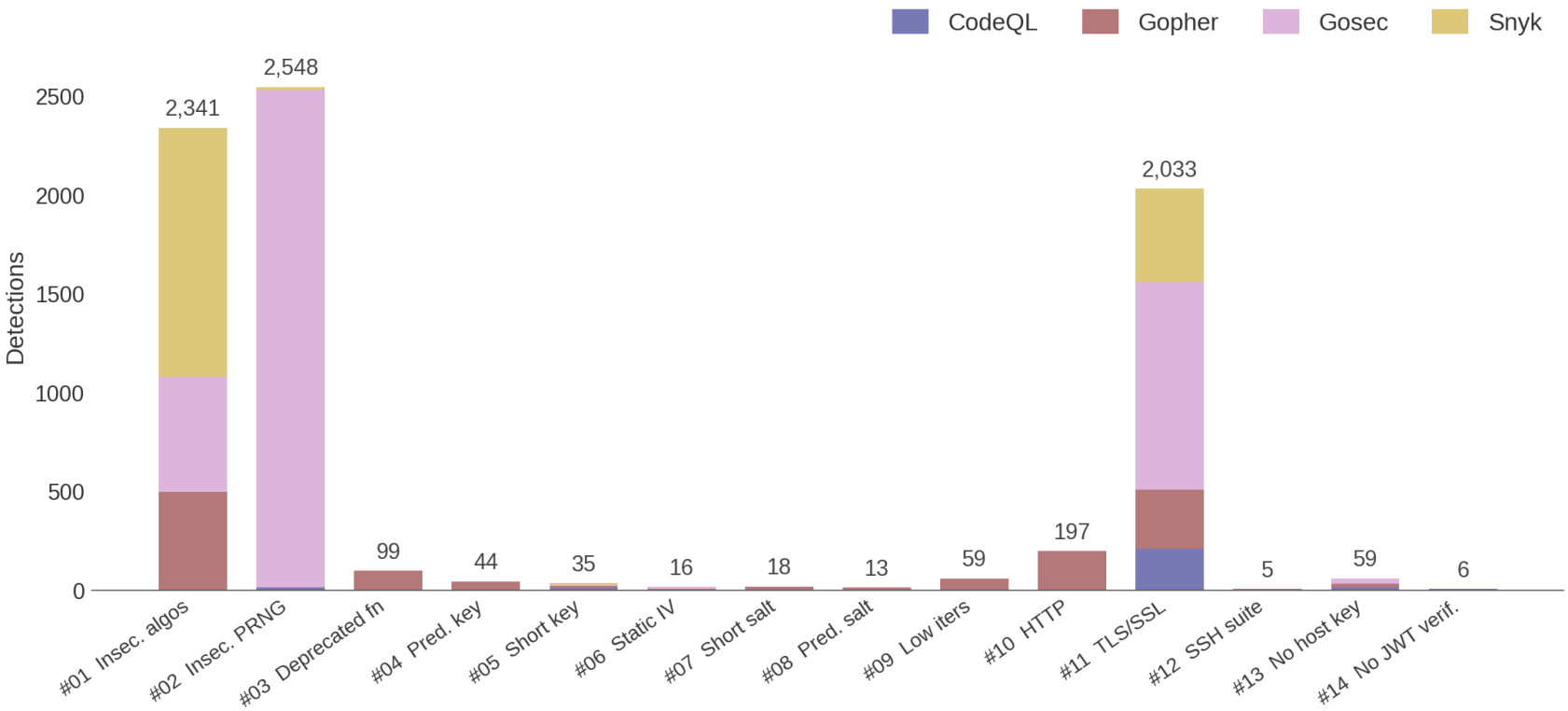
	CodeQL	Gopher	Gosec	Snyk
Projects analyzed (%)	91.8	77.1	100	100
With findings (% of analyzed)	30.9	64.8	84.5	92.7
With findings (% of dataset)	28.4	50.0	84.5	92.7

Coverage

Snyk Code and Gosec analyze all 328 projects, while Gopher fails on 75 (runtime errors) and CodeQL on 27 (database build erros).

RQ2: Raw Findings by Tool

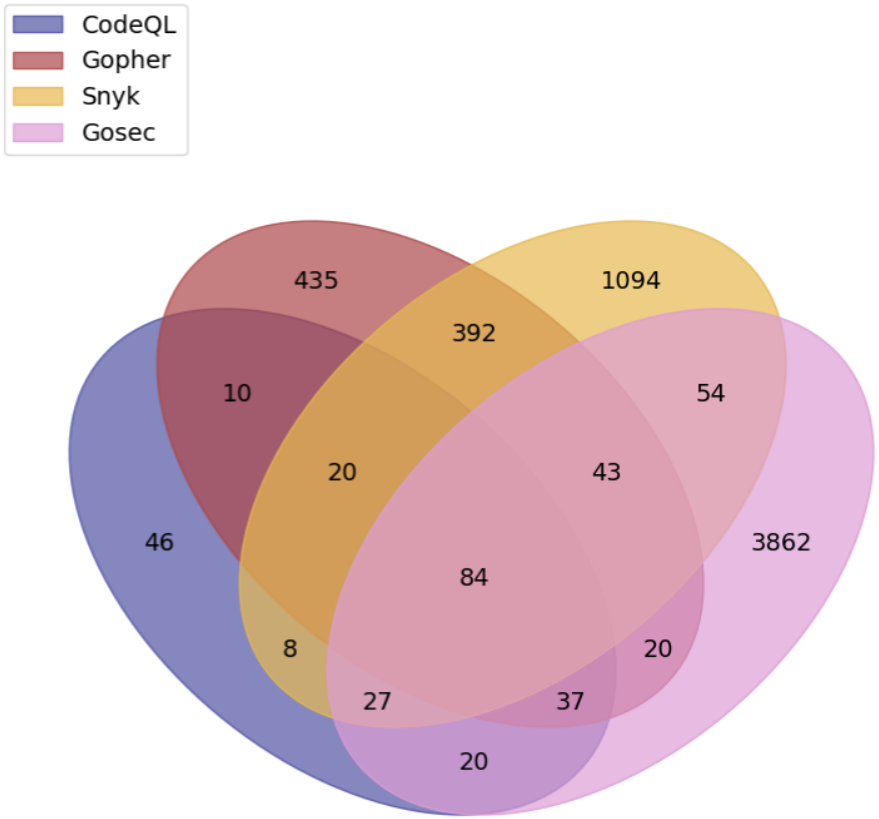
- 7,473 findings across 328 projects



- GOPHER uniquely detects issues in #04 Predictable Key
- Excessive detections indicate insensitive detection (e.g., #01, #02, #11)

RQ2: Line Level Agreement

- 6,152 unique flagged lines across the dataset.
- Only **1.3%** are flagged by **all four** tools. **11.6%** are flagged by at least **two**.



RQ2: Per-Rule Agreement

Category	#	Rule	CodeQL	Gopher	Gosec	Snyk
Primitives	01	Insecure algorithms		✓	✓	✓
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- Per-rule agreement matches on `(project, file, line, rule)`

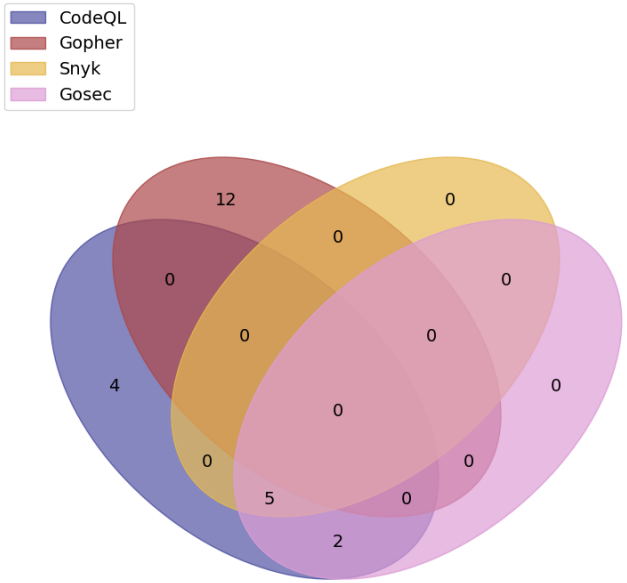
RQ2: Agreement on Rule #05

WIP

Concrete misuse: short RSA key

```
priv, err := rsa.GenerateKey(
    rand.Reader, 1024)
```

- 1024-bit RSA is below current recommendations.
- Weak keys make long-term impersonation or decryption more realistic.



Overlap. All four tools support Rule #05, but agreement is still low: they differ on whether mock/example code and broader patterns should count.

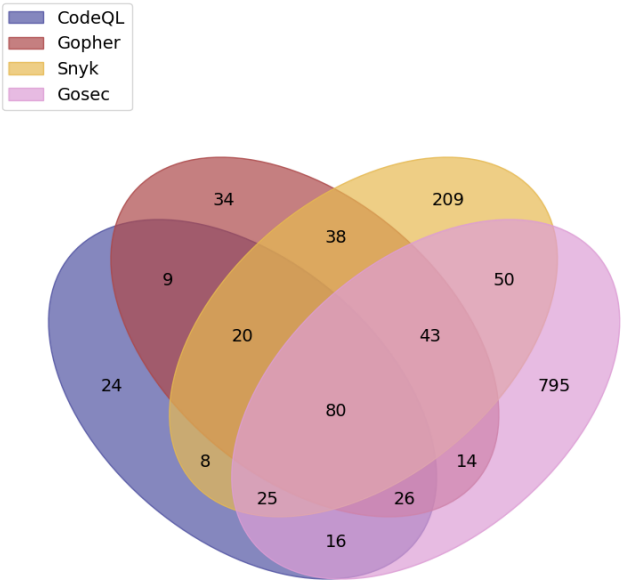
RQ2: Agreement on Rule #11

WIP

Concrete misuse: TLS certificate checks disabled

```
tls.Config{
    InsecureSkipVerify: true,
}
```

- TLS stays encrypted, but peer authentication is gone.
- A machine-in-the-middle can present its own certificate and be trusted.



Overlap. This is the largest class. All tools find true positives here, but Gosec and Snyk Code contribute many tool-unique alerts, with different precision levels.

CodeQL finds misuses the others miss

Example from `keadm batch` (Rule #13, no SSH host key check):

```
HostKeyCallback: func(...) error {
    return nil
},
```

A custom callback that accepts every host key. Only CodeQL flags this; Gosec and Gopher only match the obvious `InsecureIgnoreHostKey`.

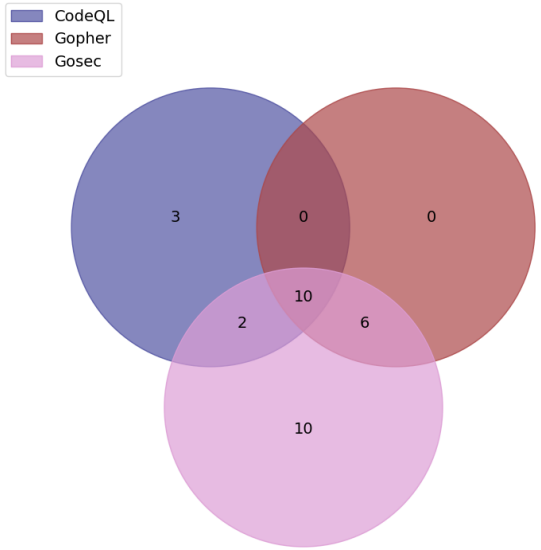
RQ2: Rule #13

WIP

Concrete misuse: SSH host key verification disabled in `keadm batch`

```
HostKeyCallback: func(...) error
    {
    return nil
},
```

- The SSH client accepts any server host key, so the remote node identity is never authenticated.
- If an attacker redirects the connection, `keadm batch` may connect to the wrong host; with password authentication, SSH credentials may be exposed.



Overlap. GOSEC primarily matches `InsecureIgnoreHostKey`. CODEQL also detects equivalent custom callbacks, so the low overlap reflects deeper semantic coverage.

Issues in Practice

WIP, maybe covered in the examples.?

1. Context-dependent usage *Rule #05, flagged by* GOPHER

```
h := blake2b.New512(nil)   // bug if used as a MAC
                           // fine for content hashing
```

Static analysis cannot distinguish security-critical vs. benign usage.

2. Pattern match misses the equivalent *Rule #13, only* CODEQL

```
cfg.HostKeyCallback = func(...) error { return nil }
```

CODEQL detects this pattern; the other tools do not report it in our study.

3. Field name flagged, value ignored *Rule #11,* GOSEC *false positive*

```
tlsConfig := &tls.Config{Certificates: []tls.Certificate{cert}, RootCAs: roots}
setTLSDefaults(tlsConfig) // MinVersion 1.2 = OK
```

GOSEC lacks data flow capabilities and flags false positives.

Implications

- The four detectors barely overlap. Only 1.3% of flagged lines are shared by all four, and 92.6% of GOSEC's findings are unique to it.
- Tools that "support" the same rule often define misuse differently. GOSEC flags 2,517 insecure PRNG cases where GOPHER flags zero, on the same code.
- For now, no single tool is reliable on its own. Running multiple detectors and treating overlap as higher-confidence is the practical strategy.

Takeaways

- We present the first systematic comparative study of cryptographic API misuse detection in Go, covering four state-of-the-art tools.
- We consolidate a shared view of the problem space: 14 misuse classes and a large-scale evaluation on 328 security-relevant open-source Go projects, yielding 7,473 detections.
- The central result is that current tools capture different slices of the problem. Their overlap is low, so no single detector gives a reliable picture on its own.

Takeaway: for Go today, combining detectors is the practical strategy, while better standardization remains an open need.